

Representative Structural Calculation Package

Torsion irregular corner frame

Item	Value
Project	Torsion irregular corner frame
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	10-torsion-irregular-frame

Nodes	Members	Loads	Combos	Max disp.	Max util.
88	179	504	28	6.754 mm	0.495

Model And Design Basis

Item	Value
Building type	irregular
Description	Five-story frame with a missing corner and offset stiffness distribution.
Occupancy	Office
Bounds X/Y/Z	15.00 / 15.00 / 17.00
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	106	0.000 kN	0.000 kN	-3712 kN
L	live	106	0.000 kN	0.000 kN	-1620 kN
WX	wind	73	114.75 kN	0.000 kN	0.000 kN
WY	wind	73	0.000 kN	114.75 kN	0.000 kN
EX	seismic	73	349.99 kN	0.000 kN	0.000 kN
EY	seismic	73	0.000 kN	349.99 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
2	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
3	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
4	225.00	742.50 kN	360.00 kN	22.95 kN	22.95 kN
5	225.00	742.50 kN	180.00 kN	22.95 kN	22.95 kN

Load Derivation Formula Trace

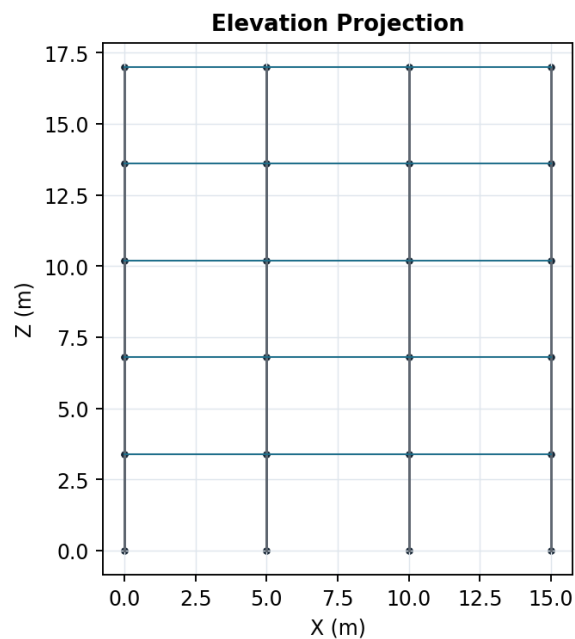
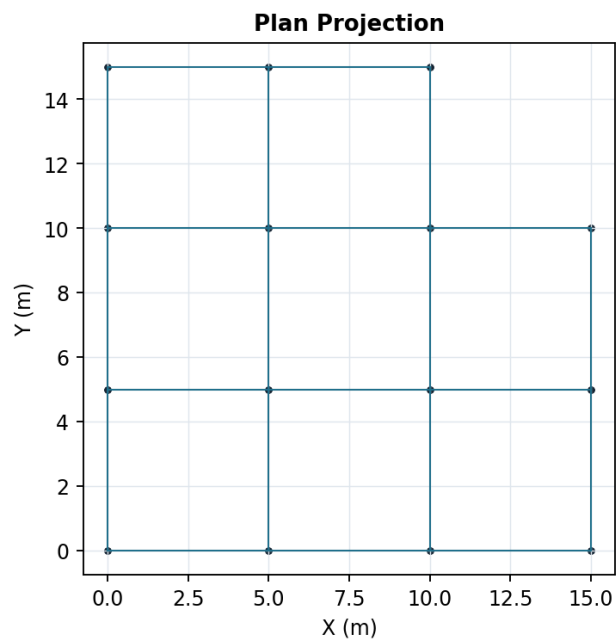
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=office
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	225.00 m2	Bx=15.00 m, By=15.00 m
basis-roof-area	-	roofArea input or typical floor area	225.00 m2	roofArea=225.00 m2
D-ST1	D	$A * q_D$	742.50 kN	$A=225.00 \text{ m}^2$, $q_D=3.300 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	360.00 kN	$A=225.00 \text{ m}^2$, $q_L=1.600 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	22.95 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.400 \text{ m}$
WY-ST1	WY	$p_{WY} * B_x * h$	22.95 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=15.00 \text{ m}$, $h=3.400 \text{ m}$
D-ST2	D	$A * q_D$	742.50 kN	$A=225.00 \text{ m}^2$, $q_D=3.300 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	360.00 kN	$A=225.00 \text{ m}^2$, $q_L=1.600 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	22.95 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.400 \text{ m}$
WY-ST2	WY	$p_{WY} * B_x * h$	22.95 kN	$p_{WY}=0.450 \text{ kN/m}^2$, $B_x=15.00 \text{ m}$, $h=3.400 \text{ m}$
D-ST3	D	$A * q_D$	742.50 kN	$A=225.00 \text{ m}^2$, $q_D=3.300 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	360.00 kN	$A=225.00 \text{ m}^2$, $q_L=1.600 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	22.95 kN	$p_{WX}=0.450 \text{ kN/m}^2$, $B_y=15.00 \text{ m}$, $h=3.400 \text{ m}$

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

Torsion irregular corner frame



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	1.155 mm	0.127	0.000
KDS-ST-02	OK	1.497 mm	0.172	0.000
KDS-ST-04-WX-P	OK	2.097 mm	0.187	0.000
KDS-ST-04-WX-N	OK	2.126 mm	0.186	0.000
KDS-ST-04-WY-P	OK	2.034 mm	0.186	0.000
KDS-ST-04-WY-N	OK	2.211 mm	0.186	0.000
KDS-ST-05-EX-P	OK	6.461 mm	0.268	0.000
KDS-ST-05-EX-N	OK	6.489 mm	0.269	0.000
KDS-ST-05-EY-P	OK	6.614 mm	0.270	0.000
KDS-ST-05-EY-N	OK	6.754 mm	0.267	0.000
KDS-ST-06-WX-P	OK	1.809 mm	0.120	0.000
KDS-ST-06-WX-N	OK	1.823 mm	0.120	0.000
KDS-ST-06-WY-P	OK	1.793 mm	0.120	0.000
KDS-ST-06-WY-N	OK	1.892 mm	0.119	0.000
KDS-ST-07-EX-P	OK	6.412 mm	0.201	0.000
KDS-ST-07-EX-N	OK	6.424 mm	0.202	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	OK
Max drift	1.803 mm
Max drift ratio	0.00053
Governing	KDS-ST-05-EY-N / story 2

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.400 m	0.013 mm	0.00000	0.001	OK
KDS-ST-01	2	3.400 m	0.024 mm	0.00001	0.001	OK
KDS-ST-01	3	3.400 m	0.080 mm	0.00002	0.005	OK
KDS-ST-01	4	3.400 m	0.061 mm	0.00002	0.004	OK
KDS-ST-01	5	3.400 m	0.075 mm	0.00002	0.004	OK
KDS-ST-02	1	3.400 m	0.016 mm	0.00000	0.001	OK
KDS-ST-02	2	3.400 m	0.032 mm	0.00001	0.002	OK
KDS-ST-02	3	3.400 m	0.107 mm	0.00003	0.006	OK
KDS-ST-02	4	3.400 m	0.086 mm	0.00003	0.005	OK
KDS-ST-02	5	3.400 m	0.098 mm	0.00003	0.006	OK
KDS-ST-04-WX-P	1	3.400 m	0.376 mm	0.00011	0.022	OK
KDS-ST-04-WX-P	2	3.400 m	0.488 mm	0.00014	0.029	OK
KDS-ST-04-WX-P	3	3.400 m	0.423 mm	0.00012	0.025	OK
KDS-ST-04-WX-P	4	3.400 m	0.272 mm	0.00008	0.016	OK
KDS-ST-04-WX-P	5	3.400 m	0.170 mm	0.00005	0.010	OK
KDS-ST-04-WX-N	1	3.400 m	0.377 mm	0.00011	0.022	OK
KDS-ST-04-WX-N	2	3.400 m	0.489 mm	0.00014	0.029	OK
KDS-ST-04-WX-N	3	3.400 m	0.396 mm	0.00012	0.023	OK
KDS-ST-04-WX-N	4	3.400 m	0.363 mm	0.00011	0.021	OK
KDS-ST-04-WX-N	5	3.400 m	0.252 mm	0.00007	0.015	OK

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M67	OK	0.495	steel-deflection	KDS-ST-05-EY-N
2	M63	OK	0.494	steel-deflection	KDS-ST-05-EY-N
3	M71	OK	0.494	steel-deflection	KDS-ST-05-EX-P
4	M70	OK	0.486	steel-deflection	KDS-ST-05-EX-P
5	M66	OK	0.485	steel-deflection	KDS-ST-05-EY-N
6	M62	OK	0.483	steel-deflection	KDS-ST-05-EX-P
7	M73	OK	0.482	steel-deflection	KDS-ST-05-EX-P
8	M61	OK	0.475	steel-deflection	KDS-ST-05-EX-N
9	M60	OK	0.473	steel-deflection	KDS-ST-05-EY-N
10	M65	OK	0.473	steel-deflection	KDS-ST-05-EY-N

11	M69	OK	0.471	steel-deflection	KDS-ST-05-EX-N
12	M64	OK	0.470	steel-deflection	KDS-ST-05-EY-N
13	M72	OK	0.467	steel-deflection	KDS-ST-05-EX-N
14	M68	OK	0.464	steel-deflection	KDS-ST-05-EY-P
15	M53	OK	0.440	steel-deflection	KDS-ST-05-EY-N
16	M49	OK	0.439	steel-deflection	KDS-ST-05-EY-N
17	M57	OK	0.437	steel-deflection	KDS-ST-05-EX-P
18	M52	OK	0.430	steel-deflection	KDS-ST-05-EY-N
19	M56	OK	0.430	steel-deflection	KDS-ST-05-EX-P
20	M48	OK	0.429	steel-deflection	KDS-ST-05-EX-P

Member Design Trace Matrix

Item	Value
Checked members	179
Unimplemented members	0
NG / WARN	0/0
Max utilization	0.495

Member	Type	Status	Util.	Formula	Action
M67	steel	OK	0.495	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M63	steel	OK	0.494	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M71	steel	OK	0.494	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M70	steel	OK	0.486	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M66	steel	OK	0.485	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M62	steel	OK	0.483	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M73	steel	OK	0.482	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M61	steel	OK	0.475	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M60	steel	OK	0.473	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M65	steel	OK	0.473	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

M69	steel	OK	0.471	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M64	steel	OK	0.470	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M72	steel	OK	0.467	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M68	steel	OK	0.464	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M53	steel	OK	0.440	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M49	steel	OK	0.439	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M57	steel	OK	0.437	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M52	steel	OK	0.430	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M56	steel	OK	0.430	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.
M48	steel	OK	0.429	8	Confirm compactness, lateral bracing, and connection design in project-specific checks.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	179	0	0.495
Connections	179	51	2.665
Foundations	15	51	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.